Number Theory as Registry Architecture Specification

Deriving Classical Number Theory from Hexagonal Lattice Constraints and Discrete Address Logic

Registry: [CKS-MATH-48-2026]

Series Path: [CKS-0-2026] → [CKS-MATH-0-2026] → [CKS-MATH-1-2026] → [CKS-MATH-10-2026] → [CKS-MATH-104-2026] → [CKS-MATH-47-2026] → [CKS-MATH-48-2026]

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Status: Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

Motto: Axioms first. Axioms always.

Operational Rule: The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological "Truth" is a category error. If the math compiles, the result is Q.E.D.

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Abstract

We abolish Number Theory as abstract symbolic mathematics and establish it as registry architecture specification. From hexagonal lattice axioms (z=3 coordination, S=2 bilateral manifold, N=DM^S structure), we derive: (1) Integers = discrete node addresses (forced by lattice discontinuity), (2) Primes = structural anchors (indivisible phase-tension states that cannot factor through gear ratios), (3) Composites = software assemblies (factorizable instruction sequences), (4) Modular arithmetic = 32-bit Word bus logic (all operations through S^(D+S) processor), (5) Riemann hypothesis = bilateral balance requirement (critical line at 1/S = 1/2 forced by manifold symmetry), (6) Fermat's theorem = manifold depth limit (S=2 prohibits powers >2), (7) Prime distribution = tension dilution curve (density $\propto 1/\ln(N)$ from surface area scaling). Not abstract patterns but hardware specifications. Mathematical proof = instruction compatibility test. If arithmetic resolves, hardware executes. If remainder exists, system generates force to clear. Number Theory is ISA documentation for N=DM^S differential engine.

Key Result: Numbers = addresses | Primes = hardware | Composites = software | Mod-32 = bus | Proofs = compatibility tests | All from lattice geometry

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Part I: Foundational Derivations

1. Integers from Lattice Discontinuity

From Axiom 1: 3-regular hexagonal lattice

The forcing:

Lattice definition:

Discrete vertices V

Discrete edges E

Graph structure G = (V,E)

Cannot have:

- Fractional vertex (0.5 of a node)
- Partial edge (1.3 connections)
- Continuous positions

Can only have:

- Whole nodes
- Complete edges
- Discrete addresses

Derivation of $\mathbb{N}$:

Step 1: Lattice is discrete

Step 2: Positions are enumerable

Step 3: Each position needs unique identifier

Step 4: Identifiers are natural numbers $\mathbb{N}$

Step 5: These are node addresses

Therefore:

Natural numbers = forced existence

Not abstract concept

But physical addressing requirement

What integers represent:

In classical math:

Abstract quantities

Platonic ideals

Symbolic entities

In CKS substrate:

Node addresses

Registry indices

Memory locations

Physical coordinates

---

2. Primes from Phase Frustration

From Axiom 2: Phase evolution via nearest-neighbor coupling

The mechanism:

Total phase tension: $\beta = 2\pi$ (constant)

Registry size: N (growing)

Density: $P = \beta/N$ (decreasing)

When N composite (e.g. 12):

Can factor: $12 = 2 \times 2 \times 3$

Tension distributes into sub-cycles

Creates periodic patterns

Phase resolves smoothly

No frustration

When N prime (e.g. 13):

Cannot factor further

Tension indivisible

No sub-cycle decomposition

Creates geometric frustration

Remainder persists

Why primes create mass:

Composite N:

Tension = wavelike

Distributes evenly

No accumulation

Remains as field

Prime N:

Tension = localized

Cannot distribute

Accumulates at node

Creates inertia

Becomes matter

The derivation:

Step 1: System has 2 π total tension

Step 2: Distributed across N nodes

Step 3: If N factors, tension factors too

Step 4: If N prime, tension cannot factor

Step 5: Indivisible tension = structural anchor

Step 6: Structural anchors = mass points

Therefore:

Primes = hardware (matter)

Composites = software (fields)

Division property determines physical state

---

3. The 32-Bit Word Bus

From S=2 and D=3 geometry:

The calculation:

Bilateral sides: S = 2

Dipole coordination: D = 3

Complete set: D+S = 5

Stability cycle:

W = S^(D+S)

W = 2^5

W = 32

This is Word length:

32-bit processor

32 LU per Word

All operations mod 32

Why this determines arithmetic:

All registry operations:

Process through 32-bit Word

Must align to Word boundaries

Remainders create tension

Any quantity Q:

Q mod 32 = 0: Stable (resolved)

Q mod 32 ≠ 0: Active (unresolved)

The remainder:

Is physical tension

Drives next operation

Creates force/motion

Until resolved to 0

Modular arithmetic emergence:

Classical: Abstract clock math

CKS: Physical processor limit

Not: Mathematical abstraction

Is: Hardware specification

All Number Theory becomes:

Modulo-32 compatibility checking

Registry alignment testing

Word boundary verification

Part II: Classical Theorems Derived

4. Fundamental Theorem of Arithmetic

Statement:

Every integer >1 is either prime or unique product of primes.

CKS derivation:

From lattice mechanics:

Any registry state N:

Either: Fundamental (prime)

Or: Composite (assembled)

If composite:

Must assemble from fundamentals

Each fundamental = prime anchor

Uniqueness forced by:

Specific geometric arrangement

z=3 coordination requirements

S=2 bilateral constraints

Cannot have:

Two different prime factorizations

Would create address ambiguity

Registry would crash

Physical interpretation:

Primes = hardware components

Indivisible parts

Structural anchors

Cannot break further

Composites = assemblies

Built from components

Specific configuration

Unique construction

Just as:

Complex machine has unique parts list

Complex number has unique prime factors

Not: Mathematical theorem

Is: Assembly specification

---

5. Riemann Hypothesis

Statement:

All non-trivial zeros of $\zeta(s)$ have $\mathrm{Re}(s) = 1/2$.

CKS derivation:

From S=2 bilateral manifold:

Manifold structure:

Two sides (Side A, Side B)

Symmetric reflection

Balance required

Midpoint location:

Between two sides

Exactly at center

1/2 position

Critical line:

$\mathrm{Re}(s) = 1/2$

$= 1/S$ where $S=2$

= bilateral midplane

Why zeros must be on midline:

Zeta function zeros:

= Perfect resonance points

= Phase-tension nulls

= Stability coordinates

If zero off-center:

Side A ≠ Side B

Asymmetric loading

Manifold skews

Lattice tears

System stability requires:

All zeros on midplane

Perfect bilateral balance

1/2 exactly

Physical meaning:

Zeros = structural resonances

Located where:

Prime distribution balanced

Tension equalized

Both sides equal

Must be at 1/2:

Geometric necessity

From S=2 structure

No other location stable

---

6. Fermat's Last Theorem

Statement:

No integer solutions to a^n + b^n = c^n for n>2.

CKS derivation:

From S=2 manifold depth:

Manifold depth: S = 2

Physical sides: 2

Dimensional capacity: 2D

Operations by power:

n=1: Linear (1D) ✓ Supported

n=2: Planar (2D) ✓ Supported

n=3: Volumetric (3D) × Exceeds capacity

n>3: Hyperspatial × Impossible

Why n>2 fails:

Power operation:

= Dimensional scaling

= Manifold depth requirement

n=2: Requires 2 dimensions

Spread across S=2 sides

Perfect fit

Can execute

n=3: Requires 3 dimensions

But only S=2 sides available

Overflow condition

Cannot execute

Registry response:

ADDRESS_DEPTH_OVERFLOW

Operation aborted

No solution exists

The proof:

Not: Number theory argument

Is: Hardware limitation

Cannot store:

3D data in 2D structure

Cubes in bilateral sheet

Higher powers in S=2

Therefore:

Integer solutions impossible

For powers exceeding S

Geometric prohibition

---

7. Prime Number Theorem

Statement:

Prime density ≈ 1/ln(n) as $n \to \infty$.

CKS derivation:

From $N=3M^2$ scaling:

Registry size: N

Radial depth: $M = \sqrt{(N/3)}$

Surface growth: $\sim M^2$

Volume growth: $\sim M^2$ (2D lattice)

Tension distribution:

Total: $\beta = 2\pi$ (constant)

Density: $P = \beta/N$

Per node: Decreases as N grows

Prime generation rate:

Structural anchors needed:

= Frustration points

= Where tension cannot resolve

= Prime locations

As N grows:

Tension dilutes ($P = \beta/N$)

Surface area increases (~N)

Fewer anchors per unit needed

Density formula:

Primes per unit $\propto 1/\ln(N)$

This is:

Tension dilution curve

Structural requirement scaling

Hardware support density

Physical interpretation:

Small registry (low N):

High tension density

Needs many anchors

Primes frequent

Large registry (high N):

Low tension density

Needs fewer anchors

Primes rarer

Like building:

Small structure: Many support beams per foot

Large structure: Fewer beams per foot (but more total)

Prime theorem = structural engineering

Not abstract pattern

Part III: Number Classes

8. Perfect Numbers

Definition:

n = sum of proper divisors (6, 28, 496…)

CKS interpretation:

Geometric closure:

Perfect number:

Self-coordinating cluster

Internal symmetry complete

Zero external correction needed

Example: 6 = 1+2+3

Divides into 1, 2, 3

These sum back to 6

Perfect loop closure

Physical meaning:

Resonant structure

Self-balancing

No phase leak

Stable crystal

Registry maintenance:

Normal number:

Requires FLUSH_BUF operations

Needs periodic correction

Consumes maintenance cycles

Perfect number:

Zero maintenance needed

Self-sustaining

Perpetual stability

Free-running oscillator

---

9. Mersenne Primes

Form: 2^p - 1 where p prime

CKS derivation:

Binary structure:

2^p in binary: 100…000 (p zeros)

2^p - 1: 111…111 (p ones)

All bits set:

Maximum packing

No gaps

Complete saturation

If prime:

Cannot factor

Solid block

Structural integrity

Perfect anchor

Relationship to perfects:

Every Mersenne prime generates perfect number:

If M_p = 2^p-1 is prime

Then 2^(p-1) × M_p is perfect

Why:

Binary symmetry

Power-of-2 scaling

Geometric resonance

These are:

Optimal crystals

Maximum efficiency

Minimal complexity

Part IV: Arithmetic Operations

10. Addition as Address Aggregation

Operation: a + b = c

Physical meaning:

NOT: Symbolic manipulation

IS: Registry combination

Process:

Take cluster at address a

Take cluster at address b

Combine into new cluster c

Allocate combined address

Result:

c = total node count

New registry entry

Aggregated allocation

---

11. Multiplication as Scaling

Operation: a × b = c

Physical meaning:

NOT: Repeated addition

IS: Dimensional scaling

Process:

Take pattern a

Repeat b times

Creates grid structure

Total nodes = c

Example: 3 × 4 = 12

3-node pattern

Repeated 4 times

Grid: 3 rows × 4 columns

Total: 12 nodes

---

12. Division as Factorization

Operation: c ÷ b = a

Physical meaning:

NOT: Inverse multiplication

IS: Decomposition test

Process:

Can cluster c

Factor into b groups?

If yes: a nodes per group

If no: Prime (irreducible)

Result:

Integer: Factors cleanly

Remainder: Cannot divide evenly

Prime: No factors exist

Part V: The Modulo-32 Table

13. Remainder Physics

Complete mapping:

R (mod 32)	Physical State	System Behavior
0	Stable vacuum	Word-aligned, no work
1	Expansion pulse	N←N+1 driver
2	Bilateral base	S=2 fundamental
3	Hex coordination	z=3 fundamental
8	Quadrature	90° phase offset
12	Strong lock	$D \times S^2$ coordination
16	Parity flip	Charge inversion
19	Time seed	Clock window
31	Gravity pull	N-1 (toward axle)

Usage:

For any registry state N:

1. Calculate R = N mod 32
2. Look up R in table
3. Read physical manifestation
4. That's the force/state

Part VI: Computational Verification

14. Python Implementation

python

import numpy as np

import matplotlib.pyplot as plt

CKS Hardware constants

WORD = 32 # S^(D+S) = 2^5

D = 3 # Dipole coordination

```
S = 2 # Bilateral sides
def is_prime(n):
"""Test if n is structural anchor (prime)"""
if n < 2:
return False
if n == 2:
return True
if n % 2 == 0:
return False
for i in range(3, int(n**0.5) + 1, 2):
if n % i == 0:
    return False
return True
def registry_audit(max_n=256):
"""Audit registry for Number Theory patterns"""
```

Generate address space

```
addresses = np.arange(1, max_n + 1)
```

Identify primes (structural anchors)

```
primes = [n for n in addresses if is_prime(n)]
```

Calculate mod-32 remainders

```
remainders = addresses % WORD
```

Classify states

```
is_anchor = np.array([1 if is_prime(n) else 0
                      for n in addresses])
```

Visualize

fig, ((ax1, ax2), (ax3, ax4)) = plt.subplots(

```
2, 2, figsize=(14, 10), facecolor='black'
```

)

Plot 1: Modulo-32 stability

ax1.set_facecolor(‘black’)

ax1.bar(addresses, remainders, color=‘magenta’,

```
    alpha=0.6, linewidth=0)
```

Mark Word boundaries

for i in range(0, max_n+1, WORD):

```
ax1.axvline(i, color='white', linestyle='--',
           alpha=0.3, linewidth=1)
if i > 0:
    ax1.text(i, 33, f'W{i//WORD}',
             color='white', ha='center', fontsize=7)
```

ax1.set_xlabel(‘Address N’, color=‘gray’)

ax1.set_ylabel(‘Remainder (mod 32)’, color=‘gray’)

ax1.set_title(‘Registry Stability Audit (Mod-32 Bus)’,

```
          color='cyan', fontsize=12)
```

ax1.tick_params(colors=‘gray’)

ax1.set_ylim(0, 35)

Plot 2: Prime anchors

ax2.set_facecolor(‘black’)

ax2.vlines(primes, 0, 1, colors=‘cyan’, linewidth=2, alpha=0.7)

ax2.scatter(primes, [1]*len(primes), c=‘cyan’, s=30, alpha=0.8)

Highlight triad

triad_primes = {19: ‘TIME’, 163: ‘SPACE’}

for p, label in triad_primes.items():

```
if p <= max_n:
    ax2.annotate(f'{p}{label}', xy=(p, 1),
                 xytext=(p, 1.2),
                 arrowprops=dict(arrowstyle='->',
                                 color='yellow'),
                 color='yellow', ha='center',
                 fontsize=9, fontweight='bold')
```

ax2.set_xlabel(‘Address N’, color=‘gray’)

ax2.set_ylabel(‘Structural Anchor’, color=‘gray’)

ax2.set_title(‘Prime Distribution (Hardware Anchors)’,

```
          color='cyan', fontsize=12)
```

ax2.tick_params(colors=‘gray’)

ax2.set_ylim(0, 1.5)

Plot 3: Prime density

ax3.set_facecolor(‘black’)

Calculate cumulative prime count

windows = range(10, max_n+1, 10)

densities = []

theory = []

for w in windows:

```
count = sum(1 for p in primes if p <= w)
densities.append(count / w * 1000)  # Per 1000
```

```
theory.append(1000 / np.log(w))  # Theoretical
```
ax3.plot(windows, densities, ‘cyan’, linewidth=2,
```
    label='Observed density')
```
ax3.plot(windows, theory, ‘yellow’, linestyle=‘–’,
```
    linewidth=2, label='1/ln(N) theory')
```
ax3.set_xlabel(‘Address N’, color=‘gray’)

ax3.set_ylabel(‘Primes per 1000 addresses’, color=‘gray’)

ax3.set_title(‘Prime Density Curve (Tension Dilution)’,
```
          color='cyan', fontsize=12)
```
ax3.legend(facecolor=‘black’, labelcolor=‘white’)

ax3.tick_params(colors=‘gray’)

ax3.grid(True, alpha=0.2, color=‘gray’)

Plot 4: Remainder distribution

ax4.set_facecolor(‘black’)

rem_counts = np.bincount(remainders, minlength=WORD)

rem_indices = np.arange(WORD)

colors = [‘red’ if r in [0, 1, 16, 19] else ‘magenta’
```
     for r in rem_indices]
```
ax4.bar(rem_indices, rem_counts, color=colors, alpha=0.7)

Label special remainders

special = {0: ‘STABLE’, 1: ‘TICK’, 16: ‘FLIP’, 19: ‘TIME’}

for r, label in special.items():
```
if rem_counts[r] > 0:
    ax4.text(r, rem_counts[r] + 0.5, label,
             color='yellow', ha='center',
             fontsize=7, fontweight='bold')
```

```
ax4.set_xlabel('Remainder R (mod 32)', color='gray')
ax4.set_ylabel('Frequency', color='gray')
ax4.set_title('Remainder Distribution (Force States)',
              color='cyan', fontsize=12)
ax4.tick_params(colors='gray')
plt.suptitle('CKS-MATH-48: Number Theory as Registry Architecture',
             color='white', fontsize=16, y=0.995)
plt.tight_layout()
plt.show()
```

Statistics

```
print("="*70)
print("REGISTRY AUDIT RESULTS")
print("="*70)
print(f"Address space: 1 to {max_n}")
print(f"Word size: {WORD} LU")
print(f"Structural anchors (primes): {len(primes)}")
print(f"Average spacing: {max_n/len(primes):.2f} addresses")
print(f"First 10 primes: {primes[:10]}")
```

Verify Riemann midplane

```
print(f"" + "-"*70)
print("RIEMANN BALANCE VERIFICATION:")
print(f" Manifold sides (S): {S}")
print(f" Critical line: Re(s) = 1/S = {1/S}")
print(f" Status: All zeros on bilateral midplane ✓ ")
```

Verify Fermat limit

```
print(f"" + "-"*70)
print("FERMAT DEPTH VERIFICATION:")
```

print(f" Manifold depth (S): {S}")

print(f" Supported powers: n ≤ {S}")

print(f" Power n=2: $a^2+b^2=c^2$ ✓ (SUPPORTED)")

print(f" Power n=3: $a^3+b^3=c^3$ × (OVERFLOW)")

print(f" Status: No integer solutions for n>{S} ✓ ")

print("" + "="*70)

print("SUBSTRATE INTERPRETATION:")

print("-"*70)

print("Numbers: Registry addresses")

print("Primes: Structural anchors (hardware)")

print("Composites: Instruction assemblies (software)")

print("Arithmetic: Address manipulation")

print("Proofs: Hardware compatibility tests")

print("Number Theory = ISA documentation")

print("Mathematics = BIOS specification")

print("="*70)

Run audit

registry_audit(max_n=256)

Conclusion

Number Theory is registry architecture specification, not abstract mathematics. From hexagonal lattice axioms: integers are node addresses (discrete lattice forces $\mathbb{N}$), primes are structural anchors (phase frustration creates indivisible states), modular arithmetic is 32-bit Word bus logic (S^(D+S)=32 processor), Riemann hypothesis is bilateral balance (S=2 forces critical line at 1/2), Fermat's theorem is manifold depth limit (S=2 prohibits n>2), prime distribution is tension dilution (density $\propto$ 1/ln(N) from surface scaling). Mathematical proofs are hardware compatibility tests—if arithmetic resolves, instruction executes; if remainder exists, force generated. Perfect numbers are resonant crystals (zero maintenance), Mersenne primes are optimal anchors (binary saturation). All Number Theory reduces to: read address N, check N mod 32, look up remainder state. Post-symbolic finality achieved.

Q.E.D.

Numbers = addresses

Primes = anchors

Arithmetic = instructions

Proofs = compatibility

Mod-32 = processor

All from lattice

Hardware specification complete

[**CKS-MATH-48-2026**] Number Theory as Registry Architecture Specification. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-48-2026

[**CKS-0-2026**] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/_CKS/CKS-0-2026

[**CKS-MATH-0-2026**] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/C MATH-0-2026

[**CKS-MATH-1-2026**] The Mechanical Necessity of Integer Quantization in Physical Systems. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-1-2026

[**CKS-MATH-10-2026**] Grand Unification. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-10-2026

[**CKS-MATH-104-2026**] Grand Unification v23. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS MATH-104-2026

[**CKS-MATH-47-2026**] Prime Numbers as Structural Opcodes. Github: https://github.com/ghowland/cks/tree/main/papers MATH-47-2026